NIHVA

NIHVA Technologies Pvt. Ltd.

2025



Design Data

CLIENT	MALAYIL BUILDERS
END CLIENT	ISRO
SITE	ALUVA
LOCATION	KERALA
NO.	001
DOCUMENT NO.	NTPL-261-001-00

Rev No.	Date	Description	Prepared by	Checked by	Approved by
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1. Objective

This document is to prove the design consideration and design calculations are in line with requirements and competent to Indian Standards.

2. Technical Details:

SNO	Description	Value	Units	Remarks
1	Door Panels			
2	Clear Opening – Width	10.3 (total Available 14)	Meters	
3	Clear Opening – Height	5	Meters	
4	Each Door Size	3.70 W x 5.00 H	Meters	
5	Type of Door	Motorized & Multi Leaf		
6	No of Leafs	4 Nos. (4 one side stacking - Either Side)		
7	Door Closing time	3 min (180 Sec)		
8	Cladding sheet	0.8 mm		
8	Material Inside door	NA		

3. Technical Details:

A. Door Construction

The door leaf frames shall be designed to withstand as per required wind speed / wind load. The frames shall be fabricated out of heavy-duty mild steel sections / profiles. The frames shall be of modular design and all sections shall be pre-fabricated at works, ready for assembly at site.

B. Protective Coating

After fabrication and before transportation all mild steel members shall be cleaned and one coat of primer applied. Before final assembly & installation at site. All bolts, nuts, washers shall be galvanized. Primer and Paint is synthetic enamel (Two Coat).

C. Cladding Sheet

0.8mm Precoated corrugated Sheet.

D. Vision Panel

No vision provided.

E. Door Movement

Each door leaf frame shall be supported at the bottom on two (2) load bearing wheels. The wheels are made of steel specially machined to run on bottom rail track. The wheels are fitted with double row sealed ball bearings. All wheels are bolted to the door frame and can be easily removed for maintenance purpose. All the door leaves are guided at the top with three (3) heavy-duty steel ball bearing rollers.

F. Bottom Tracks

The bottom tracks shall comprise of required runs of steel rails mounted on transverse support and alignment bearers along with fixing bolts etc. The bottom track assembly shall be assembled and aligned at site before grouting. The bottom tracks shall extend up to the extremities of the door-parking bay on one sides of the hangar building.

G. Top Guide Tracks

The top guides shall comprise of required sets of guide ways mounted on transverse support to the hangar structure. The door support shall be on brackets and dependent on the hangar building. The top guide tracks shall extend up to the extremities of the door-parking bay on both sides. Parking bay is excluded in the quote.

H. Buffers

Heavy-duty buffers shall be provided at the parking bay extremities of each bottom track. These buffers shall act as emergency mechanical stoppers in case of the doors overshooting their limits.

I. Drive Arrangement

A 3-phase 415 Volts, AC, 50 Hz electric motor of suitable rating shall drive each master leaf. We shall use helical bevel gear motor which has better efficiency. The drive transmission between the motor and the drive wheel shall be through an efficient gearbox. Each leaf shall be master leaf.

J. Electrical Operation

The electrical operation of each door section shall be carried out from a control panel mounted on each side on indoor structure. The operator shall have to keep the OPEN or CLOSE press button control pressed to enable the door to operate. At each extremity the door section shall automatically STOP, as the control power shall be cut-off through proximity switches. An EMERGENCY STOP BUTTON (STAY-PUT TYPE) shall also be provided on the control panel.

K. Manual Operation

In case of power failure, the doors can be operated manually by disengaging the gearbox and manually operating the gear drive with the handle provided separately. The sequence of manual operation shall be same as the electric operations.

L. Electrical Control System

General

The electrical control system shall be able to control all aspects of the door movements. The control system for each door section shall consist of a control panel, proximity switches, trailing / bunching cables etc.

Electrical Power Supply

The incoming power supply to the door control panel shall be 415V, 3-phase, 50 Hz, 4 – wire & earth. Power supply to each master leaf shall be by means of a looped trailing / bunching cable system (Festoon cale carrier system & conducting), mounted in the top guide track area from a junction box placed at the top extremities inside the hangar.

Control Panel & System

Door control panel shall consist of a VFD system mounted on panel, control fuse holders and fuses, indication lamps, interlock contactors, proximity switches on the door, etc.

Door Operation Safety

The following features shall be incorporated in the control system of the doors:

- The control panel shall be located (Left / Right) for easy operation and enable the operator to have a full view of the area during operation of the door.
- The OPEN & CLOSE Mobile Push Buttons and the EMERGENCY STOP push button is STAY PUT type
- Emergency proximity switch shall be provided at the extreme limits of travel just before the mechanical buffers. These emergency switches shall totally cut-off the control / power supply to the system in case the doors over shoot their normal limits.
- Audio / Visual Alarm – Sliding door control panel (Inside / outside) shall be provided with a Flashing Light cum Sounder that shall start & continue to operate during the time the door is in operation.
- Key Operation – An operating key switch shall be provided on each control panel to prevent un-authorized operation of the doors.

4. Design Calculation

A. Introduction

This report presents design calculations for the telescopic hanger door system, as per our discussions.

B. Design Requirements

Door Size: 3.7-meter (W) X 5-meter (H)

Material: Steel

Operation: Motorized

Safety Factors: 1.5 (static), 2.0 (dynamic)

C. Calculation

I. Material selection

Description	Profile	Material
Door construction + Steel Material	**ISMC 125 / 100**	**MS**
Top guide track + Steel Material	**ISWB150 / 120**	**MS**
Bottom track	**Rail track**	**MS**
Bottom track wheel	**Dia 310mm**	**MS**

II. Bearing Selection

Given Data

Mass of one door	= 1100 kg
Total mass of door	= 1100 x 4 door = 4400 kg
Door Closing distance	= 10300mm (10.3M)
Time to Close door	= 3 min = 180 sec (as per RFQ)
Travel Speed Required	= 10.3/180 = 0.057M/min (Approx 3.4M/Min)
Acceleration	= 0.1 m/sec^2
Acceleration due to gravity	= 9.81 m/sec^2
Coefficient of friction	= 0.1= μ

Dynamic Load Calculation

F_D **= 4571.6 N (4.5716 kN)**

Static Load Calculation

LS **= 43164 N (43.164 kN)**

Bearing Selection

No. of Wheels	8 Nos.
No. of bearing in each wheel	2 Nos.
Total no of Bearings	16 Nos.

According to static load and dynamic load factor below bearing no is suitable

MAKE – SKF

EARING PART NO – NJ 308 ECP

Dimensions		Performance	
Bore diameter	40 mm	Basic dynamic load rating	93 kN
Outside diameter	90 mm	Basic static load rating	78 kN
Width	23 mm	Reference speed	8 000 r/min
		Limiting speed	9 500 r/min
		SKF performance class	SKF Explorer

III. Motor Selection

Given data:

Mass of one door	= 1100 kg
Total mass of door	= 1100 x 4 door = 4400 kg
Door Closing distance	= 10300mm (10.3M)
Time to Close door	= 3 min = 180 sec (as per RFQ)
Travel Speed Required	+ 10.3/180 = 0.057M/min (Approx 3.4M/Min)
Acceleration	= 0.1 m/sec^2
Acceleration due to gravity	= 9.81 m/sec^2
Coefficient of friction	= 0.2 = μ

Calculate output speed (n) rpm:

Speed of the Door	= 3.4 M/min
Dia of the Roller OD	= 310 mm Dia = 0.31 M
Circumference	= 3.14 x D = 3.14*0.31 = 0.9739 M
RPM at door wheel	= 3.4/0.9739 = 3.49 RPM = ~ 3.5 RPM

Calculate Torque

T	= Total force X Wheel radius
Total force	= **4572 N**
Wheel radius	= 155 mm = 0.155 m
T	= 4572 *0.155 = 708.66N-m = ~709 N-m

Power Requirement:

POWER = 2*π*N*T / 60 Watts
= 2*3.14*3.5*709/ (60*0.7) =371.04 W=0.37104kW
=0.72115*2 (with FoS 2) = 0.742 kW

Motor Make:

Suitable as per power requirement

IV. Wind Force on Panels

Geological study at client side: -

client location	- Aluva Kerala
wind speed	- 39 m/sec
Temperature	– 42^0C
Humidity	– 98.1 %
Air density	– 1.2 kg/m^3

Wind pressure: -

P=0.6 × V^2 – as per IS 875 - Part3
P=0.6*39^2
P=912.6 Pa

Wind load: -

F= P x A

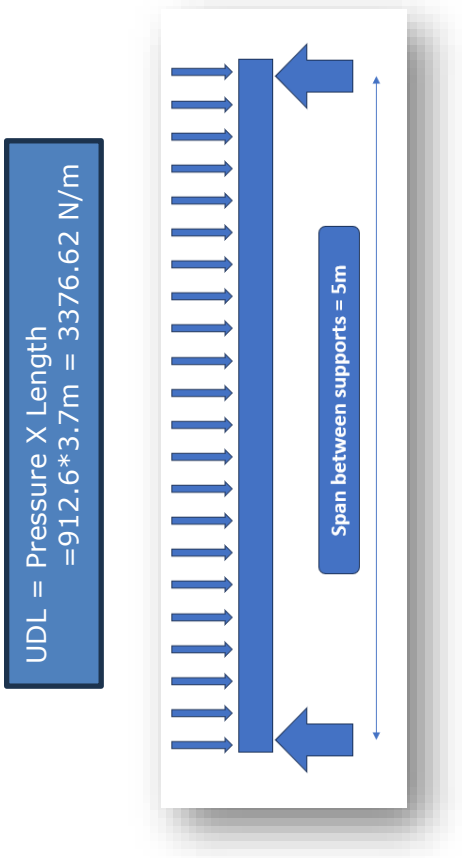
A = Area of door leaf
= 3.7 x 5
=18.5 m^2

F =P x A
= 912.6x18.5
= 16883.1 N
Consider factor of safety is 2
So, wind load = 16883.1 x 2

Wind load per door = **33766.2 N**

Reaction of Door

Total Area 3.7 x 5 meters
Pressure 912.6 N/m^2



Total Area 3.7 x 5 meters
Pressure 912.6 N/m^2

Ra = wL/2 and Rb = wL/2
Ra = Rb = wL/2 = 3376.62N/m *5 / 2 = 8441.55N (860.5 Kg)

Guide Roller bearings
Roller QTY per door = 6 Nos

Load on each roller = 8441.55 / 6 = 1406.925 N

Bearing Selected is 6203zz – ID 17 x OD 40 x 12 THK

Dimensions		Performance	
Bore diameter	17 mm	Basic dynamic load rating	9.95 kN
Outside diameter	40 mm	Basic static load rating	4.75 kN
Width	12 mm	Reference speed	38 000 r/min
		Limiting speed	24 000 r/min
		SKF performance class	SKF Explorer